

SUPERCONDUCTIVITY

Q1. The critical field of niobium is 10^5 A/m at 8K and 2×10^5 A/m at 0K. calculate the critical temperature of the element.

Given: - $T = 8K$, $H_c(0) = 2 \times 10^5$ A/m, $H_c(T) = 10^5$ A/m

Formula: - $H_c(T) = H_c(0) \left(1 - \frac{T^2}{T_c^2}\right)$

Solution: -

$$T = T_c \left[1 - \frac{H_c(T)}{H_c(0)}\right]^{1/2}$$

$$T_c = 8 \times \left[1 - \frac{10^5}{2 \times 10^5}\right]^{-1/2}$$

$$T_c = 11.3137^\circ \text{ K}$$

Ans: - critical temperature is 11.3137° K

Q2. The transition temperature for Pb is 7.2K. however at 5K it loses the superconducting property if subjected to magnetic field of 3.3×10^4 A/m. Find the maximum value of H which will allow the metal to retain its superconductivity at 0K.

Given: - $T_c = 7.2K$, $H_c(T) = 3.3 \times 10^4$ A/m, $T = 5K$

Formula: - $H_c(T) = H_c(0) \left(1 - \frac{T^2}{T_c^2}\right)$

Solution: - $H_c(0) = H_c(T) \left(1 - \frac{T^2}{T_c^2}\right)^{-1}$

$$= 3.3 \times 10^4 \left(1 - \frac{5^2}{7.2^2}\right)^{-1}$$
$$= 6.3737 \times 10^4 \text{ A/m}$$

Ans: - the critical field at 0K is 6.3737×10^4 A/m

Q3. The transition temperature for lead is 7.26K. The maximum critical field for the material is $8 \times 10^5 \text{ A/m}$. Lead has to be used as a superconductor subjected to a magnetic field of $4 \times 10^4 \text{ A/m}$. What precaution will have to be taken?

Given: -

$$T_C = 7.26\text{K}, \quad H_C(0) = 8 \times 10^5 \text{ A/m}, \quad H_c(T) = 4 \times 10^4 \text{ A/m}$$

Formula: - $H_c(T) = H_c(0) \left(1 - \frac{T^2}{T_c^2}\right)$

Solution: -
$$T = T_c \left[1 - \frac{H_c(T)}{H_c(0)}\right]^{1/2}$$
$$= 7.26 \left[1 - \frac{4 \times 10^4}{8 \times 10^5}\right]^{1/2}$$
$$= 7.0761$$

Ans: - The temperature of lead should be maintained below 7.0761K above will lose its superconductivity.

Q4. A superconductor has a critical temperature 3.7K at zero magnetic field. At 0K, the critical magnetic field is 0.0306 Tesla. What is the critical magnetic field at temperature 2K?

Given: - $T_C = 3.7\text{K}, \quad H_C(0) = 0.0306\text{T}, \quad T = 2\text{K}$

Formula: - $H_c(T) = H_c(0) \left(1 - \frac{T^2}{T_c^2}\right)$

Solution: - $H_c(T) = 0.0306 \left(1 - \frac{2^2}{3.7^2}\right) = 0.0217 \text{ T}$

Ans: - The critical magnetic field is 0.0217 T